

Math 110 Fall 2026 Running Review Document

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1 Disclaimer

This review document will be based on the lectures of Alan Hammond in the Fall of 2026 (our course). These lectures are themselves based on the lectures of Edward Frenkel, and Sheldon Axler's *Linear Algebra Done Right*. These are supplementary and are not a replacement for the lecture notes, or for attending lecture. They are meant as a short recap with some guiding questions based on my personal understanding of the content.

2 Super-Short Test Review

Fields: What are the field axioms, and what are some common fields? You should also be able to prove some basic properties using the field axioms, like that if $ab = 0$, either $a = 0$ or $b = 0$. Make sure to know $\mathbb{F}_2$.

Vector Spaces: What are the vector space axioms and how do they differ from the field axioms? You should focus on the operations and their properties. Again, some basic properties from the axioms include

things like $0v = \vec{0}$ for every v in the vector space. Why does the field that the vector space is over matter? You should be able to show that common things are vector spaces, like $\mathbb{F}^n$, the space of sequences over $\mathbb{F}$, the polynomials, and the matrices.

Subspaces: What is the definition of a subspace, and what tool do we use to make it easier to show something is a subspace? You should know some examples of subspaces and be able to determine whether a given space is or is not a subspace.

Subspace Sums: Why do we take sums instead of unions? What is special about the subspace sum? What are direct sums and how do they sharpen this definition? You should be able to compute sums, look at examples of why this subspace sum is not an invertible operation/doesn't behave like vector sums, and given $U \subset \mathbb{R}^n$ you should be able to find V so $U \oplus V = \mathbb{R}^n$.

Linear Independence and Span: You should be able to define linear independence and use that definition to show that a set of vector is linearly independent. You should be able to define a span, find spanning sets for some vector spaces, and both prove and use the deletion lemma. Combining the two ideas gives a basis, and you should be able to work with bases, do computations with them, and use the building up and cutting down lemmas to manipulate bases.

Dimension:

Linear Maps:

3 Sets and Proofs

A set, in its essence, is just some collection of objects. That is the definition we use in discrete math, and for many purposes, though it is a naive definition (Renee Descarte and several others brought up paradoxes in this basic idea, leading to the creation of the Zermelo-Fraenkel axioms as the foundation of set theory). Most of the set theory we need is a review of Math 55, Discrete mathematics. You should be comfortable using set notation, like saying $x \in M$ for “ x is an element of M ”, or using set-builder notation like $\{(x, x^2) \mid x \in \mathbb{R}\} \subset \mathbb{R}^2$ (*what is this set?*). You should also recognize unions ($A \cup B$), intersections ($A \cap B$), complements (A^C), set differences ($A \setminus B$), and cartesian products ($A \times B$). This is also a good time to review the basics of proofs and generally interesting spaces, like the natural numbers $\mathbb{N}$, the integers $\mathbb{Z}$, and the real numbers $\mathbb{R}$.

1. Prove that $(\bigcup_{a \in A} U_a)^C = \bigcap_{a \in A} (U_a^C)$. (This is DeMorgan's law)
2. Define injective, surjective, and bijective. Construct a function which is injective but not surjective. Construct another that is surjective but not injective.
3. Construct a function $f : \mathbb{N} \rightarrow \mathbb{Z}$ to show that these two sets have the same cardinality (i.e. construct a bijective function).

Throughout this course, we will be using proofs. Proofs are a formal argument that gives evidence for some mathematical statement. The following is an example of such a proof presented during lecture.

Example: x^2 is even if and only if x is even.

Proof: Assume first that x is even. Then, $x = 2k$ for some integer k , and $x^2 = (2k)^2 = 4(k^2) = 2(2k^2)$. Since $m = 2k^2$ is also an integer, $x^2 = 2m$ is even. This proves the statement “If x is even, then x^2 is even”.

Since we claimed an “if and only if”, we also need to prove the claim “if x^2 is even, then x is even”. We do this by proving the *contrapositive*, that if x is not even (odd), then x^2 is not even.

Indeed, if x is not even, $x = 2k + 1$ for some integer k . Then, $x^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1 = 2m + 1$ for $m = 2k^2 + 2k$ an integer. Therefore, x^2 also fails to be even.

The essence of our proof here is to expand on the definition in our assumption for each case, then use some other structure or calculation to show that some criteria follows from this definition. We will do this very often. One important method of proof for our course is called *proof by induction*. Mathematical induction

(not electrical) is the concept that if we want to prove some condition R_n for every number $0, 1, 2, 3, 4, 5, \dots$, we first show that R_0 holds (our “base case”), then show that if R_n holds, then R_{n+1} does ($R_n \Rightarrow R_{n+1}$). A great example is the following:

Example: For every nonnegative integer n ,

$$\sum_{i=0}^n i = \frac{n(n+1)}{2}$$

First, when $n = 0$, the sum is empty and so clearly both sides are 0. Next, assume that this holds for n . Then,

$$\begin{aligned} \sum_{i=0}^{n+1} i &= (n+1) + \sum_{i=0}^n i = (n+1) + \frac{n(n+1)}{2} \\ &= \frac{2(n+1) + n(n+1)}{2} = \frac{(n+1)(n+2)}{2} \end{aligned}$$

so that the condition holds for $n+1$. Therefore, by the principal of mathematical induction, we have the claimed statement.

You should practice writing proofs, as they are the main mode of communicating mathematical facts.

1. Show that for all non-negative integers n ,

$$\sum_{i=0}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

2. Prove that $\mathbb{Z} \times \mathbb{Z}$ has the same cardinality as $\mathbb{Z}$.
3. Prove that there are infinitely many prime numbers. *Hint:* Assume that there are finitely many primes $p_1, \dots, p_n$. Can you find a prime not in this list? If so, you have found a *contradiction* and your assumption must be false.

4 Fields

Fields are a special type of set with very nice addition and multiplication. In particular, they have structure built to generalize the properties of the real numbers. Fields are a *unital ring* in which every nonzero element has a multiplicative inverse and multiplication is commutative. Let’s break down this definition. The “ring” part means that addition and multiplication are defined on the elements of the ring (where addition is commutative and associative, and multiplication is associative and distributive). This addition also has an additive identity 0 so $0 + r = r$ for a ring element r , as well as an additive inverse $-r$ so $r + (-r) = 0$. Next, “unital” means we have a multiplicative identity 1 so $1(r) = r$ for every element r of our field F , and the multiplicative inverse means for each r that is not zero, we have $\frac{1}{r}$ so $r \frac{1}{r} = 1$. These fields are very important in abstract algebra, so it is worth looking over the full axiomatic definition a few times to understand it. We will mostly be working over the fields $\mathbb{R}$ and $\mathbb{C}$ (the complex numbers), though we will retain some abstraction. The complex numbers are particularly important because they are what is called an *algebraically closed* field, which just means that every polynomial with complex numbers as coefficients has all of its roots in the complex numbers. This fact is called the Fundamental Theorem of Algebra (a proof of it often relies on a set of theory called *complex analysis*, which is a bit beyond this course). You should recall how to multiply, add, conjugate, and divide by complex numbers.

- Which of the following are fields? $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{R} \times \mathbb{R}$ (is multiplication defined here?), $\{0, 1\}$

- Consider $\mathbb{Z}_3$, which is the set $\{0, 1, 2\}$ where addition and multiplication are defined modulo 3 (so $1 + 2 = 0$, $2 + 2 = 1$, $1 + 1 = 2$). Is this a field?
- Let $a = 1 + 3i$ and $b = 4 + 5i$. Compute

$$\frac{1}{\bar{a}b}$$

- Show

$$\left(\frac{-1 + i\sqrt{3}}{2}\right)^3 = 1$$

- Show $0a = 0$ for all $a \in \mathbb{F}$. *Hint:* $0 = 0 + 0$
- Show that if $1 \in \mathbb{F}$ has additive inverse -1 , then $(-1)a = -a$ for all $a \in \mathbb{F}$.

The “standard” structure we will use from this point on and compare many things to is $\mathbb{F}^n$, where $\mathbb{F}$ is a field. This is the set of n -tuples or lists $\{(a_1, \dots, a_n) \mid a_i \in \mathbb{F}\}$. Addition is defined component-wise, so

$$(a_1, \dots, a_n) + (b_1, \dots, b_n) = (a_1 + b_1, \dots, a_n + b_n)$$

and we define a *scalar multiplication* by elements of the field: for $\alpha \in \mathbb{F}$,

$$\alpha(a_1, \dots, a_n) = (\alpha a_1, \dots, \alpha a_n)$$

1. What is $\mathbb{R}^3$?
2. Show that this multiplication is distributive, i.e. $\lambda(a + b) = \lambda a + \lambda b$ for $\lambda \in \mathbb{F}$, and $a, b \in \mathbb{F}^n$.
3. Show $1x = x$ for $x \in \mathbb{F}^n$.
4. For $x \in \mathbb{F}$, show that $x^n = x(x)(x)\dots(x) = 0$ for some n if and only if $x = 0$.

Remark: The reason we bother with this abstract definition of a field is that there are many other fields, and they can provide cases that work against our intuition. For example, we have the integers modulo a prime. “modulo” is an operation that means “the remainder when dividing by”. If we are working modulo 5, we have

$$\begin{aligned} 14 + 27 &= 41 = 1 + 6(5) \equiv 1 \pmod{5} \\ 2 * 4 &= 8 = 3 + 1(5) \equiv 3 \pmod{5} \\ 5 &= 0 + 1(5) \equiv 0 \pmod{5} \\ 4 &\equiv 4 \pmod{4} \\ 4 &= (-1) + 1(5) \equiv -1 \pmod{5} \end{aligned}$$

The last three cases show that working modulo 5 only requires the numbers $\{0, 1, 2, 3, 4\}$. We call the integers mod 5 by the name $\mathbb{Z}_5$. This is also a field (check the axioms, e.g. $2 * 3 \equiv 1 \pmod{5}$ gives a multiplicative inverse for 2) that is important in a field of math called abstract algebra.

Examples to Know: $\mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathbb{Z}_2$.

5 Vector Spaces

5.1 Definition

A vector space is a broader abstract concept of a space with “additive structure”, but the multiplicative structure is through *scalar* multiplication and so the space isn’t quite as nice as just a ring. Our prototypical vector space is the $\mathbb{F}^n$ we defined above.

For any vector space, we begin with some set V and define two operations on the set. The first is addition $+: V \times V \rightarrow V$, which is defined to be commutative and associative, have an additive identity element 0 , and have for each $v \in V$ an inverse $-v$ so $v + (-v) = 0$. Next, any vector space is associated with a field $\mathbb{F}$, and we call V a vector space over the field $\mathbb{F}$. This gives the second operation, scalar multiplication $*: \mathbb{F} \times V \rightarrow V$ which is defined to be associative and distributive, while we also assume $1 * v = v$ for every vector $v \in V$.

You should note that the additive identity 0 is unique, so there cannot be two distinct vectors with the additive identity property.

In the case of $\mathbb{F}^n$, the addition was component-wise addition of vectors and the multiplication was component-wise multiplication by the field element. Therefore, all of these properties follow from the field axioms. In the future, we will actually show that many vector spaces (the finite dimensional ones) look like (are isomorphic to) our example $\mathbb{F}^n$, which is why we keep bringing this example up.

1. Using the vector space axioms, show that the set of 2×2 matrices whose elements are real numbers, $M_2(\mathbb{R})$, is a vector space.
2. Is $\mathbb{R}$ a vector space over $\mathbb{C}$? Is $\mathbb{C}$ a vector space over $\mathbb{R}$?
3. Like for fields, show $0v = 0$ for every $v \in V$ and $(-1)v = -v$ for all $v \in V$.

Examples to Know: $\mathbb{F}^n$, $\mathbb{F}^S = \{\text{functions } f: S \rightarrow \mathbb{F}\}$, the polynomials $P = \{a_0 + a_1x + \dots + a_nx^n \mid a_i \in \mathbb{F}\}$, Matrices, Sequences $S_{\mathbb{N}} = \{(a_0, a_1, \dots) \mid a_i \in \mathbb{F}\}$.

Vector spaces are the foundation of our understanding of a “linear space” where we want to study linear structure. This structure seems weird and abstract at first because it is- it took many mathematicians working on affine geometry before such definitions were made. While Rene Descartes and Pierre de Fermat began identifying system solutions with plane curves as early as 1636, removing coordinates from these processes (i.e. using abstract objects) first began with Bolzano in 1804, and the modern definition of the vector space was first adopted by Giuseppe Peano in 1888 based on the work of Cayley on matrices (1857) and Grassmann’s abstraction of addition and scalar multiplication (1844).

5.2 Subspaces

If we consider a vector space like $\mathbb{R}^3$, we may also notice that things which look like other vector spaces are contained in $\mathbb{R}^3$. For example, $\{(0, y, z) \mid y, z \in \mathbb{R}\}$ looks very much like $\mathbb{R}^2$. This brings us to the concept of a *subspace*. A subset U of a vector space V is a subspace if it is a vector space with the operations inherited from V . This means that, when checking to see if a subset is a subspace, we don’t need to check whether the addition is associative or the multiplication is distributive, we just need to check that these operations are defined on U . In particular, we need to check that for all $u, v \in U$ and $a, b \in \mathbb{F}$, $au + bv \in U$.

1. Prove that, given $U \subset V$ satisfies this criterion, that U is a vector space.
2. Given a 2×2 matrix A , show that the solution set

$$\{x \in \mathbb{R}^2 \mid Ax = 0\}$$

is a subspace. Show that this fact is not true in general if we look at x solving $Ax = [1, 0]$.

Examples to Know: lines, planes, and other spaces in $\mathbb{R}^n$. Solutions to $Ax = 0$ for a matrix A . Polynomials of degree at-most n . Special functions, like continuous or differentiable functions. Special matrices like upper triangular or orthogonal matrices (will talk about more later in the class). Recurrence subspaces $R = \{(a_1, a_2, \dots) \mid a_n = \sum_{i=1}^m c_i a_{n-i} \text{ for } n \geq m\} \subset S_{\mathbb{N}}$.

5.3 Sums

This definition is quite broad and gives many more subspaces than the simple form above. From Calc 3, you may remember that all planes and lines in $\mathbb{R}^3$ are subspaces. Further, if we consider the space of all functions from $[0, 1] \rightarrow \mathbb{R}$ with pointwise addition and scalar multiplication, denoted $\mathbb{R}^{[0,1]}$, the continuous functions are a subspace and the differentiable functions are a subspace of the continuous functions (and so a subspace of the original vector space).

There are many other ways to produce subspaces. For example, the intersection of subspaces is also a subspace. This is not true in general for unions: consider $A = \{(x, y) \mid y = 0\} \cup \{(x, y) \mid x = 0\}$. This is the union of the x and y axis in $\mathbb{R}^2$. Each component of the union is a subspace, but $(1, 0) + (0, 1) \notin A$ shows that A is not closed under addition.

A better way to form a larger subspace is to instead take the sum of subspaces. Given $V_1, \dots, V_m \subset V$ that are subspaces,

$$V_1 + \dots + V_m = \{v_1 + \dots + v_m \mid v_i \in V_i \text{ for all } i\}$$

In particular, the sum is the *smallest* subspace containing each of the constituent subspaces, which is to say that if $W \subset V$ is a subspace satisfying $V_i \subset W$ for all i , then $V_1 + \dots + V_m \subset W$.

1. What is $\{(x, y) \mid y = 0\} + \{(x, y) \mid x = 0\}$?
2. Prove that the intersection of subspaces is still a subspace.
3. Consider the real vector space $R = \{(a_0, a_1, a_2, \dots) \mid a_i \in \mathbb{R}, \sum |a_i| < \infty\}$. Show $R_i = \{(a_0, a_1, \dots) \mid a_j = 0 \text{ for } j \neq i\}$ is a subspace. Find vectors $v_i \in R_i$ so $\sum_{i=1}^{\infty} v_i \notin R$, and so we cannot extend the above definition to infinite sums without some care.
4. Prove that the union of two subspaces is a subspace if and only if one subspace contains the other.

Recall that we define $\mathbb{F}^n$ using *coordinates* $(a_1, \dots, a_n)$. This is a very special case of a sum where $\mathbb{F}^n = \sum \mathbb{F}_i$ for

$$\mathbb{F}_i = \{(0, \dots, 0, a_i, 0, \dots, 0) \mid a_i \in \mathbb{F}\}$$

The special part is that these subspaces don't have a "large" overlap, which means that any two people looking at the same point will write the same coordinates with respect to this sum.

For example, if we add

$$\{(0, 0, z, a) \mid z, a \in \mathbb{R}\} + \{(x, y, 0, 0) \mid x, y \in \mathbb{R}\}$$

we get all of $\mathbb{R}^4$, but we could just as well add

$$\{(x, y, 0, 0)\} + \{(0, y, z, a)\}$$

The way we can symbolically notice the overlap in the subspaces is that, in the first case, we may decompose any vector

$$(x, y, z, a) = (x, y, 0, 0) + (0, 0, z, a)$$

as the sum of two vectors from the two subspaces, and there is only one possible way to do this. In the second case

$$(x, y, z, 0) = (x, y - 1, 0, 0) + (0, 1, z, 0) = (x, y - 2, 0, 0) + (0, 2, z, 0) = \dots$$

The overlapping parts of the vector space mean that we can take infinitely many different decompositions.

This leads us to the definition of a *direct sum*. Suppose $V_1, \dots, V_m$ are subspaces of V . The sum $V_1 + \dots + V_m$ is called a direct sum if each element $v \in V_1 + \dots + V_m$ may be written uniquely as the sum $v = v_1 + \dots + v_m$ for $v_i \in V_i$. In this case, we denote the sum $V_1 \oplus \dots \oplus V_m$.

In particular, $U + W$ is direct if and only if $U \cap W = \{0\}$.

1. If $V_1, V_2, W \subset V$ are subspaces such that $V = V_1 \oplus W = V_2 \oplus W$, is it true that $V_1 = V_2$ (try experimenting in $\mathbb{R}^2$)?
2. Consider $\Lambda = f : \mathbb{R}^2 \rightarrow \mathbb{R}$. Let $A = \{f \mid f(x, y) = -f(y, x)\}$ and $S = \{f \mid f(x, y) = f(y, x)\}$. Show $\Lambda = A \oplus S$ (we will look at this example again much later).
3. Consider $W = \{(x, y, x, y) \in \mathbb{R}^4\}$. Find a subspace V so $W \oplus V = \mathbb{R}^4$.

6 Linear Independence and Span

Starting from any vector $v \in V$, we may immediately make a subspace by simply taking $\{cv \mid c \in \mathbb{F}\}$ (this is a line in V). Any subspace containing v must also contain this since it is closed under scalar multiplication, so this is the smallest subspace containing v . What if we take two vectors $v_1 \neq v_2$? To make a similar subspace, we must include scalar products of both vectors, as well as the sums of scalar products (all linear combinations). Thus, we define the span

$$\text{span}\{v_1, \dots, v_n\} = \{\lambda_1 v_1 + \dots + \lambda_n v_n \mid \lambda_i \in \mathbb{F}\}$$

to be the set of all finite linear combinations, which is again the smallest subspace containing the vectors (the word finite is used here because, if we take infinitely many vectors to build the span, we still only allow finite sums in the span). How big are these “smallest subspaces”?

We could just try counting the vectors, but we have a strange possibility that can throw a wrench in understanding their size. When $v_n \in \text{span}\{v_1, \dots, v_{n-1}\}$, $\text{span}\{v_1, \dots, v_n\} = \text{span}\{v_1, \dots, v_{n-1}\}$. To exclude this case, we call a set of vectors $\{v_1, \dots, v_n\}$ linearly independent when $v_i \notin \text{span}\{v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_n\}$ for any i , or equivalently when

$$\lambda_1 v_1 + \dots + \lambda_n v_n = 0$$

implies $\lambda_1 = \dots = \lambda_n = 0$. (In the infinite-vector case, we just claim this holds for all finite subcollections of vectors).

In the case that $\{v_1, \dots, v_n\}$ are linearly independent, we say their span has dimension n . Further, if $V = \text{span}\{v_1, \dots, v_n\}$ and the v_i are linearly independent, we call them a *basis* of V (because we can fully describe V by this nice span). In particular, every $w \in V$ may be written uniquely as $w = c_1 v_1 + \dots + c_n v_n$, and we will later call these $c_1, \dots, c_n$ the coordinates with respect to this basis.

If vectors are not linearly independent, we call them linearly dependent. If $\{v_1, \dots, v_m\}$ are linearly dependent, one of the vectors is in the span of the other by definition. Therefore, we may shave away unnecessary vectors until we arrive at a subcollection with the same span but which are linearly independent. In other words, every collection of vectors spanning V contains a basis. Alternately, if we start with a linearly independent collection of vectors $\{v_1, \dots, v_m\}$ in V which do not span V , we can certainly add vectors from V not in their span. This means the new collection $\{v_1, \dots, v_{m+1}\}$ will be linearly independent. If we keep doing this, we can fill out V and eventually obtain a basis. These are sometimes referred to as the “cutting down” and “building up” lemmas. They allow us to say things like that, in a vector space V , a linearly independent list of vectors is always at-most the length of any spanning list.

1. For a finite-dimensional vector space V , prove the “cutting down” and “building up” lemmas (in infinite dimensions, you need a tool called Zorn’s lemma).
2. Prove that, in $\mathbb{F}^n$, there is no linearly independent list of vectors of length $n + 1$ and no spanning list of length $n - 1$.
3. If $v_1, \dots, v_n$ are a basis for V , is $\{w_1, \dots, w_n\}$ also a basis for $w_k = v_1 + \dots + v_k$? Prove your answer.
4. For the above problem, show that the v_i are a basis if and only if the w_i are.
5. Prove $\mathbb{R}^{\mathbb{R}}$ is infinite dimensional.
6. If $\{v_1, \dots, v_n\}$ and $\{w_1, \dots, w_m\}$ are two lists of linearly independent vectors, is $\{v_1 + w_1, \dots, v_n + w_n\}$ also linearly independent?

7. Given an arbitrary subspace $W \subset V$, prove that there is U so $W \oplus U = V$.
8. Given $U \oplus W = V$, and a basis $\{u_1, \dots, u_m\}$ for U , $\{w_1, \dots, w_n\}$ for W , is $\{u_1, \dots, u_m, w_1, \dots, w_n\}$ a basis for V ? Prove your answer.

Examples to Know: The standard basis $e_1, \dots, e_n$ for $\mathbb{F}^n$ ($e_1 = (1, 0, \dots, 0)$, $e_2 = (0, 1, 0, \dots, 0)$...). A basis for the polynomials by $\{1, x, x^2, \dots\}$.

6.1 Dimension

Using the building-up and cutting down lemmas above, we know that any two bases of a finite-dimensional vector space must have the same length (verify this). We call this length the dimension of a vector space V , denoted $\dim(V)$. Using the same lemmas, we know that the dimension respects inclusion $\dim(U) \leq \dim(V)$ if $u \subset V$, and that any linearly independent set whose length is the dimension (and any spanning set whose length is the dimension) must be a basis. For example, one can check that $(1, 1)$ and $(2, 1)$ are linearly independent, and thus are a basis for $\mathbb{R}^2$ without checking span (this will appear later when we talk about linear transformations as the fact that such a transformation is injective if and only if it is surjective).

1. Check that if U and W have $U \oplus W = V$, then $\dim(U) + \dim(W) = \dim(V)$.
2. Check more generally that if $U, W \subset V$ are subspaces, then $\dim(U) + \dim(W) - \dim(U \cap W) = \dim(U + W)$.
3. Use the above two exercises to show that for U, W subspaces of V , $U \oplus W = V$ if and only if $\dim(U) + \dim(W) = \dim(V)$ and $U + W = V$.
4. Find a formula for $\dim(U + W + S)$ for subspaces $U, W, S \subset V$ (you will need at least one sum).
5. Find a basis for the set of polynomials with coefficients in $\mathbb{R}$ of degree at-most n . What is the dimension of this subspace?
6. If U and W are subspaces of $\mathbb{F}^7$, show they must have nontrivial intersection (their intersection contains a nonzero element).

Examples to Know: You should know how to prove all of the above mentioned facts, and connect some common subspaces (of $\mathbb{F}^n$) to pictures via their dimension. The polynomial subspaces are also important here.

7 Linear Maps

A linear map $T : V \rightarrow W$ is a function between vector spaces over a field $\mathbb{F}$ which respects their operations, so $T(au + bv) = aT(u) + bT(v)$ for $a, b \in \mathbb{F}$ and $u, v \in V$. In particular, the set of linear maps from V to W , denoted $L(V, W)$, is itself a vector space when we define pointwise addition and scalar multiplication for these functions. Some common examples are

- The map $T : V \rightarrow W$ which maps all vectors to $0 \in W$
- The identity $T : V \rightarrow V$ so $T(v) = v$
- Matrices $M(n)$ acting on $\mathbb{R}^n$
- Differentiation and integration
- Multiplication by a field element (scalar multiplication)
- $T : S_{\mathbb{R}} \rightarrow S_{\mathbb{R}}$ which takes $T(a_1, a_2, \dots) = (a_2, a_3, \dots)$ (this is the Bernoulli Shift).

We can add even more structure to $L(V, W)$ when we define the product of $T \in L(U, V)$ and $S \in L(V, W)$ to be the composition $ST = S \circ T$, since this is associative and distributive.

1. Prove that if $T \in L(V, W)$, then $T(0) = 0$.

2. Is the map $T : \mathbb{P} \rightarrow \mathbb{R}^2$ defined by

$$T(p(x)) = (p(0), \int_0^1 p(x)dx, p'(0))$$

linear?

3. Let $\{v_1, \dots, v_n\}$ be a basis for V and $\{w_1, \dots, w_m\}$ be a basis for W with $m \geq n$. Construct a linear map $T : V \rightarrow W$ which takes $T(v_i) = w_i$ for $i = 1, \dots, n$. *Hint:* If this map exists, what is $T(av_1 + bv_2)$?

4. Consider the map $\phi : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $\phi(x, y) = x + 1$. Is it homogeneous? Is it additive? Answer the same questions for $\psi : \mathbb{R}^2 \rightarrow \mathbb{R}$ given by $\psi(x, y) = \sqrt{x^2 + y^2}$. What does this show you about these two properties we use to define linearity?

5. Is the map $T : \mathbb{P} \rightarrow \mathbb{P}$ given by $T(p(x)) = q(p(x))$ for some $q \in \mathbb{P}$ a linear map?

6. Suppose V is a finite dimensional vector space and $T \in L(V)$ is such that $ST = TS$ for every $S \in L(V)$. Prove T is a scalar multiple of the identity (a *homotethy*). *Hint:* Consider the two cases where $\dim(V) = 1$ and $\dim(V) > 1$. In the first case, show every linear map must be a homotethy. In the second case, show that if $T \neq \alpha I$, then there always exists S so $ST \neq TS$.

As with many functions, we consider the case of injective and surjective linear maps. In particular, the linear structure means that we generate some interesting subspaces when looking into these.

First, for any $w \in W$ and $T \in L(V, W)$, and any v_0 so $T(v_0) = w$,

$$\{v \in V \mid T(v) = w\} = v_0 + \{v \in V \mid T(v) = 0\}$$

so that T is injective if and only if second set is precisely $\{0\}$. We give this set a name, the *null space* or the *kernel* $Null(T) = \{v \in V \mid T(v) = 0\}$. You may prove that the null space is actually a subspace, and we will talk a lot more about it when it comes to quotient spaces.

Similarly, the image or range of the linear map is also a subspace, and we denote it $Im(T) = \{Tv \mid v \in V\}$. As with normal functions, T is surjective if and only if this image is all of W .

However, it is useful to understand these two subspaces because of the Rank-Nullity Theorem: if V is a finite-dimensional vector space and $T \in L(V, W)$, then

$$\dim(V) = \dim(Null(T)) + \dim(Im(T))$$

(i.e. the dimension is the nullity plus the rank) This is an important proof to know, and so we sketch it here. Begin by establishing a basis $\{u_1, \dots, u_m\}$ for $Null(T)$, then extend it to a basis for V $\{u_1, \dots, u_m, v_1, \dots, v_n\}$. Notice that $Im(T)$ is spanned by $\{T(u_1), \dots, T(u_m), T(v_1), \dots, T(v_n)\}$, but of course the first m vectors here are 0 so it is spanned by $\{T(v_1), \dots, T(v_n)\}$. Furthermore, these vectors are linearly independent: if $\alpha_1 T(v_1) + \dots + \alpha_n T(v_n) = 0$, then

$$T(\sum \alpha_i v_i) = 0$$

and so $\sum \alpha_i v_i \in Null(T)$. By the basis assumption, we must then have that $\sum \alpha_i v_i = 0$, and so $\alpha_i = 0$ since the v_i are linearly independent. Therefore, $\dim(Im(T)) = n$, $\dim(Null(T)) = m$, and $\dim(V) = m + n$ as we claimed.

Some remarkable results come from this theorem. For example, $T \in L(V)$ is injective if and only if it is surjective (and so it is bijective and invertible if and only if it is injective). If $T \in L(V, W)$ and $\dim(V) > \dim(W)$, then T must fail to be injective. If instead $\dim(V) < \dim(W)$, it must fail to be surjective. This is also how we prove that a homogeneous system of linear equations has nonzero solutions!

1. Is the derivative from $\mathbb{P} \rightarrow \mathbb{P}$ injective? Is it surjective?

2. Prove the facts claimed in the last paragraph above.

3. Let V have odd dimension. Show there is no $T \in L(V)$ so $\text{Im}(T) = \text{Null}(T)$. If W has even dimension, give an example of $S \in L(W)$ so $\text{Im}(S) = \text{Null}(S)$.
4. Show that the set of injective linear maps is not a subspace of $L(V, W)$.
5. Prove the converse of some of the facts claimed in the last paragraph: if $\dim(V) \leq \dim(W)$, then there exists an injective linear map from V to W . Similarly, if $\dim(V) \geq \dim(W)$, there exists a surjective map.
6. Let V be infinite-dimensional. Show that for any $T \in L(V, W)$, either $\text{Im}(T)$ or $\text{Null}(T)$ must be infinite-dimensional.
7. Given $T : V \rightarrow W$ and $S : W \rightarrow U$, show that if ST is injective, T must be injective. Furthermore, show

$$\text{Null}(ST) = T^{-1}(\text{Null}(S))$$

and

$$\dim(\text{Null}(ST)) \leq \dim(\text{Null}(S)) + \dim(\text{Null}(T))$$

with equality if and only if $\text{Null}(S) \subset \text{Im}(T)$.

8. (Sided inverses) Let $T : V \rightarrow W$ be linear. Show that T is surjective if and only if there exists $S \in L(W, V)$ so TS is the identity. Show that T is injective if and only if there is $S \in L(W, V)$ so ST is the identity.
9. There are a ton of problems here and a lot more in Axler's chapter on this- pick 5 from his chapter that don't repeat information from the above (29 is cool)

Examples to Know: There aren't many affixed examples here, but you should be able to come up with various simple examples of injective, non-injective, surjective, non-surjective, linear, and non-linear maps. Projection maps and inclusion maps are often quite helpful.

7.1 Matrices

When V and W are both finite-dimensional vector spaces, it is easy to represent a linear map $T : V \rightarrow W$ in an information-dense format called a matrix. In particular, we establish a pair of bases $\{v_1, \dots, v_m\}$ for V and $\{w_1, \dots, w_n\}$ for W . Then, we may write every $v \in V$ as $v = \sum_{i=1}^m a_i(v)v_i$ for some scalars $a_i(v) \in \mathbb{F}$, and we may similarly decompose W . In this situation, we can give v the "name" or coordinates $(a_1, \dots, a_m)$.

Let $T(v_i) = \sum_{j=1}^n b_{i,j}w_j$. Notice that for any $v = (a_1, \dots, a_m)$,

$$\begin{aligned} T(v) &= \sum_{i=1}^m a_i T(v_i) = \sum_{i=1}^m \sum_{j=1}^n a_i b_{i,j} w_j \\ &= \sum_{j=1}^n \left[\sum_{i=1}^m a_i b_{i,j} \right] w_j \end{aligned}$$

so in coordinates on W , this is $([\sum_{i=1}^m a_i b_{i,1}], \dots, [\sum_{i=1}^m a_i b_{i,m}])$. You may recall from a previous linear algebra class that this is precisely the vector generated by

$$[b_{i,j}] \begin{bmatrix} a_1 \\ \vdots \\ a_m \end{bmatrix}$$

and so we can fully encode the action of T by this matrix $[b_{i,j}]$ for scalars $b_{i,j}$ (what size is this matrix?). We denote this matrix as $\mathcal{M}(T)$. Notice that the sum and scalar product of matrices is the same as that of the linear operations:

$$\mathcal{M}(\lambda T + S) = \lambda \mathcal{M}(T) + \mathcal{M}(S)$$

whereas matrix multiplication is precisely composition

$$\mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(ST)$$

(justifying why we call it the product of linear maps). You should be able to prove these two facts by evaluating on vectors. For example, let $\mathcal{M}(S) = [a_{i,j}]$ be a $p \times m$ matrix (why do we need m columns?). Then,

$$[\mathcal{M}(S)\mathcal{M}(T)]_{j,k} = \sum_{r=1}^m a_{j,r} b_{r,k}$$

and so

$$\mathcal{M}(S)\mathcal{M}(T) \begin{bmatrix} c_1 \\ \vdots \\ c_m \end{bmatrix} = \begin{bmatrix} \sum_{k=1}^n \sum_{r=1}^n a_{1,r} b_{r,k} c_k \\ \vdots \\ \sum_{k=1}^n \sum_{r=1}^n a_{p,r} b_{r,k} c_k \end{bmatrix}$$

and you should get the same when looking at the composition.

This matrix depends quite heavily on the choice of bases involved. If we call the basis $\{v_1, \dots, v_m\}$ for V by $\mathcal{A}$ and $\{w_1, \dots, w_n\}$ for W by $\mathcal{B}$, then we may instead write $\mathcal{M}_{\mathcal{A}}^{\mathcal{B}}(T)$. We write the domain in the bottom so this can look like fractions canceling when we keep track of bases:

$$\mathcal{M}_{\mathcal{B}}^{\mathcal{C}}(S)\mathcal{M}_{\mathcal{A}}^{\mathcal{B}}(T) = \mathcal{M}_{\mathcal{A}}^{\mathcal{C}}(ST)$$

In general, to find the matrix for T under the bases $\mathcal{A}$ and $\mathcal{B}$, we follow the computations we did original to notice that the k th column of $\mathcal{M}_{\mathcal{A}}^{\mathcal{B}}(T)$ is

$$\begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

where

$$T(v_k) = \sum_{j=1}^n c_j w_j$$

(the coordinates of this image under $\mathcal{B}$).

One of our next big topics will look at how we can move between these coordinate representations and bases, and what that means for these matrices.

[Incomplete- Add Stochastic matrices, row/column fact?]

1. Let $T: \mathbb{P}_3 \rightarrow \mathbb{P}_2$ be given by $T(p(x)) = p'(x) + p''(x)$. Find bases for these two spaces and the matrix representation of T under these bases.
2. Consider $\mathbb{R}^2$ under the basis e_1 and e_2 , and let T be given by the matrix

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Under the alternative basis

$$\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

consider the matrix

$$\begin{bmatrix} 5/3 & 4/3 \\ -1/3 & 1/3 \end{bmatrix}$$

Does this matrix represent T ?

8 Isomorphisms

How do we know when two vector spaces are equal, and what should that mean? A copy of $\mathbb{R}^2$ sits in $\mathbb{R}^3$: $\{(x, y, 0) \mid x, y \in \mathbb{R}\}$ looks identical to $\mathbb{R}^2$ but is not strictly “equal”.

To answer this, we recall that two sets A, B with no extra structure on them have the same size if and only if there exists a bijection $f : A \rightarrow B$ between them (a function which is both injective and surjective, which is the same as being invertible, which means to have $f^{-1} : B \rightarrow A$ so $f \circ f^{-1} = Id_B$ and $f^{-1} \circ f = Id_A$). Since we are adding some additive structure and some scalar multiplication structure, equality would mean we just want these two operations to look the same on either side of the bijection, so $f(\lambda v + w) = \lambda f(v) + f(w)$. However, this is just a linear map!

Therefore, we call a linear map $T \in L(V, W)$ an isomorphism if it is invertible, or equivalently if it is bijective. In this case, the inverse is also linear. We call two vector spaces V and W isomorphic if there exists a linear map between them.

Checking an isomorphism is easier than normal in finite-dimensional vector spaces of the same dimension, since the Rank-Nullity theorem implies that a map T is invertible iff. it is injective iff. it is surjective. Recall that a linear map $T : V \rightarrow W$ is injective if and only if there exists $S \in L(W, V)$ so $TS = I$. Similarly, T is surjective if and only if there exists $S : W \rightarrow V$ so $ST = I$. Therefore, the previous claim from the Rank Nullity theorem says that when V and W have the same dimension, a left inverse exists if and only if a right inverse exists, in which case this is a two-sided inverse T^{-1} .

Notice further that invertible linear maps must take bases to bases: if $\{v_1, \dots, v_n\}$ is a basis for V and $T : V \rightarrow W$ is invertible and linear, $\{T(v_1), \dots, T(v_n)\}$ is a basis for W . To see this, notice that the map being injective implies the $T(v_i)$ are linearly independent, and the map being surjective implies they span W . In particular, V and W must have the same dimension.

Given any V and W of the same dimension, we can also find a pair of bases $\{v_1, \dots, v_n\}$ and $\{w_1, \dots, w_n\}$ and construct a map so $T(v_i) = w_i$. It can be shown that this map is then injective and surjective. This means that two finite-dimensional vector spaces are isomorphic if and only if they have the same dimension (so every n -dimensional vector space over $\mathbb{F}$ is isomorphic to $\mathbb{F}^n$). For example, $L(V, W)$ is isomorphic to $\mathbb{F}^{\dim(V) \dim(W)}$, where the canonical isomorphism is by using the matrix of a linear transformation.

This revelation about bases and dimension tells us that every invertible transformation looks like a change-of-basis and every change-of-basis is just an isomorphism.

1. Given that $TU \rightarrow V$ and $S : V \rightarrow W$ are isomorphisms, check that TS is.
2. If $TS : U \rightarrow W$ is an isomorphism of finite dimensional vector spaces, show that $T : V \rightarrow W$ and $S : U \rightarrow V$ also are isomorphisms.
3. Suppose V is finite-dimensional and $S, T \in L(V, W)$. Show $\text{range}(S) = \text{range}(T)$ if and only if there exists invertible $E \in L(V)$ so $S = TE$.
4. Show that V is isomorphic to $L(\mathbb{F}, V)$.

9 Products, Direct Sums, Quotient Spaces

We have found ways to observe vector spaces, to differentiate them, and to relate some subspaces to other vector spaces. We now look at ways to produce new vector spaces and create such relationships.

The first and easiest case is that of the product. Given two vector spaces V and W over the same field $\mathbb{F}$, define

$$V \times W = \{(v, w) \mid v \in V, w \in W\}$$

with the component-wise operations $(v_1, w_1) + a(v_2, w_2) = (v_1 + av_2, w_1 + aw_2)$. This is a new vector space, with $\dim(V \times W) = \dim(V) + \dim(W)$.

One interesting application of this product is when V_1, V_2 are subspaces of a vector space V . Then, if we have $V_1 \cap V_2 = \{0\}$, the direct sum $V_1 \oplus V_2$ is isomorphic to this product $V_1 \times V_2$. This is seen in an easy way: for arbitrary subspaces W_1, W_2 , the sum $W_1 + W_2$ is direct if and only if the map $T(w_1, w_2) : W_1 \times W_2 \rightarrow W_1 + W_2$ given by $T(w_1, w_2) = w_1 + w_2$ is injective.

While it is easy to build up like this, there is a slightly more complicated way to “shrink” vector spaces. It comes from the idea of a linear map: assume we have a vector space V and subspace U . Then, if $U = \ker(T)$ for a linear map $T : V \rightarrow W$, the image of T “looks like” pushing everything in U to 0, and keeping the rest of V up to this identification. This is how we begin to construct quotient spaces.

For this definition, let V be a vector space and $U \subset V$ a subspace. For $v \in V$, define $v + U = \{v + u \mid u \in U\}$, which is called the translate of U . Notice that for $v, w \in V$, we may have two possibilities. First, $v - w \in U$, in which case $v + U = w + U$. Second, $v - w \notin U$, in which case $v + U \cap w + U = \emptyset$.

We define the quotient space V/U to be the set of translates

$$\{v + U \mid v \in V\}$$

By our observation above, these are distinct subsets partitioning V . If we further define addition and scalar multiplication

$$\begin{aligned}(v + U) + (w + U) &= (v + w) + U \\ \lambda(v + U) &= \lambda v + U\end{aligned}$$

then V/U is a vector space.

Given arbitrary V, U , we may now define a map $\pi : V \rightarrow V/U$ by $\pi(v) = v + U$. Notice that the kernel of this map is precisely U , so *every subspace is the kernel of a linear map*.

Given an arbitrary linear map $T : V \rightarrow W$, we can also construct a map $\tilde{T} : V/\ker(T) \rightarrow W$ given by

$$\tilde{T}(v + \ker(T)) = T(v)$$

which is well-defined because for any $w \in v + U$, $w = v + u$ for some $u \in U$ and $T(w) = T(v) + T(u) = T(v)$. This new map $\tilde{T}$ is also injective with the same image as T , and so it gives an isomorphism $V/\ker(T) \simeq \text{Im}(T)$. Hence, every quotient space is isomorphic to the image of a linear map. This is a big concept that will appear later in your math journey as well, and is commonly called the first isomorphism theorem.

1. Given a basis $\{v_1, \dots, v_m\}$ for V and $\{w_1, \dots, w_n\}$ for W , find a basis for $V \times W$.
2. Let $T : V \rightarrow V$ and $S : W \rightarrow W$ be given, with matrix representations A and B , let $T \times S(v, w) = (T(v), S(w))$. What is the matrix representation for $T \times S$?
3. Show that $L(V, W_1 \times W_2) \simeq L(V, W_1) \times L(V, W_2)$ as vector space.
4. Prove that $V^m \simeq L(\mathbb{F}^m, V)$ as vector spaces.
5. Prove that the translate addition and scalar multiplication are well defined: i.e. show that if $v_1 + U = v_2 + U$, then $\lambda(v_1 + U) = \lambda(v_2 + U)$ and if $w_1 + U = w_2 + U$ also, then $(v_1 + U) + (w_1 + U) = (v_2 + U) + (w_2 + U)$.
6. Let $U \subset V$ be a subspace. If $\{v_1 + U, \dots, v_m + U\}$ is a basis for V/U and $\{u_1, \dots, u_n\}$ for U , show $\{v_1, \dots, v_m, u_1, \dots, u_n\}$ is a basis for V .
7. Let $U \subset V$ be a subspaces and V/U be finite-dimensional. Prove $V \simeq U \times V/U$. *Hint:* You cannot use the above problem because U may be infinite-dimensional. However, you can still use a basis of V/U to construct a map $U \times V/U \rightarrow V$ which is injective.
8. Let $U \subset V$ be a subspace so V/U is finite-dimensional. Show there exists W so $W \oplus U = V$.

10 Duality